```
> restart;#neutrino Muon;
```

```
> de := diff(g(r), r$2) = r*g(r)/Rs;
```

$$de := \frac{d^2}{dr^2} g(r) = \frac{r\,g(r)}{Rs} \tag{1}$$

```
> ics := g( 2· Rs) = 0 , D(g)(0) = Rs;
```

$$ics := g(2\,Rs) = 0, D(g)(0) = Rs \tag{2}$$

```
> sol := dsolve({de, ics});
```

$$sol := g(r) = -\left(2\,Rs\,\pi\left(\text{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\text{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right) - \text{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\text{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right)\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\text{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right) + \text{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)3^{1/6}\right)\right) \tag{3}$$

```
> #Rs:=8.46;
```

```
> g(r) := (2 Rs π AiryBi(-2 (-1/Rs)^(1/3) Rs) AiryAi(-(-1/Rs)^(1/3) r)) /
    ((-1/Rs)^(1/3) Γ(2/3) (3^(2/3) AiryAi(-2 (-1/Rs)^(1/3) Rs) + 3^(1/6) AiryBi(
    -2 (-1/Rs)^(1/3) Rs))) - (2 AiryAi(-2 (-1/Rs)^(1/3) Rs) Rs π AiryBi(-(-1/Rs)^(1/3) r
    -2 (-1/Rs)^(1/3) Rs)));
```

$$g := r \mapsto \left(2\cdot Rs\cdot\pi\cdot\text{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot\text{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\text{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right) + \text{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{1/6}\right)\right) - \left(2\cdot\text{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot Rs\cdot\pi\cdot\text{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\text{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right) + \text{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{1/6}\right)\right) \tag{4}$$

```
> g(0);
```

$$\left(2\,Rs\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)\Big/\left(3\,\Gamma\left(\frac{2}{3}\right)^{2}\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/6}\right)\right)-\left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)Rs\,\pi\,3^{5/6}\right)$$

$$\left.\left.{}^{3}Rs\right)3^{1/6}\right)\right)$$

```
> # g(r) ·∂u(t)/∂t+u(t)· ∂g(r)/∂r·c=c;# thus for r=0 and t≠0 we
    have;
```

```
> dsolve(g(0)·diff(v(t), t) + v(t)·Rs·c - c = 0);
```

$$v(t)=\left(\frac{e^{-\frac{3\,c\,\Gamma\left(\frac{2}{3}\right)^{2}\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/6}\right)t}{2\,\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}-\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}}{Rs}+c_{1}\right)e^{-\frac{3\,c\,\Gamma\left(\frac{2}{3}\right)^{2}\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/6}\right)t}{2\,\pi\left(-\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}} \tag{6}$$

```
>   sol1 := v(t)·g(r);
```

$$sol1 := v(t)\left(\left(2\,Rs\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/}\right.\right.\right.\right.$$

$$\left.\left.{}^{3}Rs\right)3^{1/6}\right)\right)-\left(2\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)Rs\,\pi\,\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}r\right)\right)\Big/$$

$$\left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/}\right.\right.\right.$$

$$\left.\left.\left.{}^{3}Rs\right)3^{1/6}\right)\right)\right) \tag{7}$$

```
> H(r, t) := (1/Rs + e
  -(3 c Γ(2/3)^2 (-1/Rs)^(1/3) (3^(2/3) AiryAi(-2 (-1/Rs)^(1/3) Rs) + 3^(1/6) AiryBi(-2 (-1/Rs)^(1/3) Rs)) t)
   / (2 π (-AiryAi(-2 (-1/Rs)^(1/3) Rs) 3^(5/6) + AiryBi(-2 (-1/Rs)^(1/3) Rs) 3^(1/3)))

  C1) · ((2 Rs π AiryBi(-2 (-1/Rs)^(1/3) Rs) AiryAi(-(-1/Rs)^(1/3) r)) /

  ((-1/Rs)^(1/3) Γ(2/3) (3^(2/3) AiryAi(-2 (-1/Rs)^(1/3) Rs) + 3^(1/6) AiryBi(

  -2 (-1/Rs)^(1/3) Rs))) - (2 AiryAi(-2 (-1/Rs)^(1/3) Rs) Rs π AiryBi(-(-1/Rs)^(1/3) r

  -2 (-1/Rs)^(1/3) Rs)))));#_C1=C1;
```

$$H := (r, t) \mapsto \left( \frac{1}{Rs} + e^{-\frac{3 \cdot \left(3^{2/3} \cdot \mathrm{AiryAi}\left(-2 \cdot \left(-\frac{1}{Rs}\right)^{1/3} \cdot Rs\right) + \mathrm{AiryBi}\left(-2 \cdot \left(-\frac{1}{Rs}\right)^{1/3} \cdot Rs\right) \cdot 3^{1/6}\right) \cdot \left(-\frac{1}{Rs}\right)^{1/3} \cdot \Gamma\left(\frac{2}{3}\right)^2 \cdot c \cdot t}{2 \cdot \pi \cdot \left(-\mathrm{AiryAi}\left(-2 \cdot \left(-\frac{1}{Rs}\right)^{1/3} \cdot Rs\right) \cdot 3^{5/6} + \mathrm{AiryBi}\left(-2 \cdot \left(-\frac{1}{Rs}\right)^{1/3} \cdot Rs\right) \cdot 3^{1/3}\right)}} \cdot C1 \right) \cdot \left( \left(2 \cdot Rs \cdot \pi \cdot \mathrm{AiryBi}\left(-2 \cdot \left(-\frac{1}{Rs}\right)^{1/3} \cdot Rs\right) \cdot \mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} \cdot r\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\right. \right. \tag{8}$$

$$\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{1/6}\right)\right)-\left(2\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot Rs\cdot\pi\cdot\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right)\Big/\left(\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{1/6}\right)\right)$$

```
> restart; #The Redshift;Rs:=8.209331978 in Billion LY;
```

```
> eq1 := 1/c^2 diff(f(r, t), t, t) - diff(f(r, t), r, r) = 0;
```

$$eq1 := \frac{\frac{\partial^2}{\partial t^2} f(r,t)}{c^2} - \frac{\partial^2}{\partial r^2} f(r,t) = 0 \tag{9}$$

```
> iv1 := f(2· Rs, 0) = 0, D[1](f)(0, 0) = -Rs;
```

$$iv1 := f(2\,Rs, 0) = 0, \mathrm{D}_1(f)\,(0,0) = -Rs \tag{10}$$

```
> Sol := pdsolve([eq1, iv1]);
```

$$Sol := f(r,t) = \frac{\left(_c_2\, t^2 + 4\,Rs^2 - 2\,Rs\,r + 2\,c_3\, t\right) c^2 + _c_2\left(-4\,Rs^2 + r^2\right)}{2\,c^2} \tag{11}$$

```
> pdsolve([eq1, iv1])/(4·Pi·r^2);
```

$$\frac{f(r,t)}{4\,\pi\,r^2} = \frac{\left(_c_2\, t^2 + 4\,Rs^2 - 2\,Rs\,r + 2\,c_3\, t\right) c^2 + _c_2\left(-4\,Rs^2 + r^2\right)}{8\,\pi\,r^2\,c^2} \tag{12}$$

```
> f1 := subs(t = 0, %)
```

$$f1 := \frac{f(r,0)}{4\,\pi\,r^2} = \frac{\left(4\,Rs^2 - 2\,Rs\,r\right) c^2 + _c_2\left(-4\,Rs^2 + r^2\right)}{8\,\pi\,r^2\,c^2} \tag{13}$$

```
> f1 := ((4 Rs^2 - 2 Rs r) c^2 + _c_2 (-4 Rs^2 + r^2))/(8 π r^2 c^2);
```

$$f1 := \frac{\left(4\,Rs^2 - 2\,Rs\,r\right) c^2 + _c_2\left(-4\,Rs^2 + r^2\right)}{8\,\pi\,r^2\,c^2} \tag{14}$$

```
> h := series(f1, r);
```

$$h := \frac{4\,Rs^2\,c^2 - 4\,Rs^2\,_c_2}{8\,\pi\,c^2}\,r^{-2} - \frac{1}{4}\,\frac{Rs}{\pi}\,r^{-1} + \frac{_c_2}{8\,\pi\,c^2} \tag{15}$$

```
> subs(r = 2·Rs, h);
```

$$\frac{4\,Rs^2\,c^2 - 4\,Rs^2\,_c_2}{8\,\pi\,c^2}\,2\,Rs^{-2} - \frac{1}{4}\,\frac{Rs}{\pi}\,2\,Rs^{-1} + \frac{_c_2}{8\,\pi\,c^2} \tag{16}$$

```
> solve(int((4 Rs^2 c^2 - 4 _c_2 Rs^2)/(8 π c^2) ·2 ·Rs^-2 - 1/4 Rs/π ·2· Rs^-1 + _c_2/(8 π c^2), r = 0
  ..2· Rs) = 2· Rs)
```

$$\left\{Rs = Rs,\ c = c,\ _c_2 = -\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2\right\} \tag{17}$$

```
> subs(_c_2 = -8/7 π c^2 + 4/7 c^2, (4 Rs^2 c^2 - 4 _c_2 Rs^2)/(8 π c^2) r^-2 - 1/4 Rs/π r^-1
  + _c_2/(8 π c^2));
```

$$\frac{4\,Rs^2\,c^2 - 4\,Rs^2\left(-\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2\right)}{8\,\pi\,c^2\,r^2} - \frac{Rs}{4\,\pi\,r} + \frac{-\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2}{8\,\pi\,c^2} \tag{18}$$

```
> series((4 Rs^2 c^2 - 4 (-8/7 π c^2 + 4/7 c^2) Rs^2)/(8 π c^2 r^2) - Rs/(4 π r) + (-8/7 π c^2 + 4/7 c^2)/(8 π c^2), r);
```

$$\frac{4\,Rs^2\,c^2 - 4\,Rs^2\left(-\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2\right)}{8\,\pi\,c^2}\,r^{-2} - \frac{1}{4}\,\frac{Rs}{\pi}\,r^{-1} + \frac{-\frac{8}{7}\,\pi\,c^2 + \frac{4}{7}\,c^2}{8\,\pi\,c^2} \tag{19}$$

```
> simplify((4 Rs^2 c^2 - 4 (-8/7 π c^2 + 4/7 c^2) Rs^2)/(8 π c^2) r^-2 - 1/4 Rs/π r^-1
  + (-8/7 π c^2 + 4/7 c^2)/(8 π c^2));
```

(20)

$$\frac{\left(-4\,\pi+2\right) r^2-7\,Rs\,r+16\left(\pi+\frac{3}{8}\right) Rs^2}{28\,r^2\,\pi} \tag{20}$$

```
> series(((-4*Pi + 2)*r^2 - 7*Rs*r + 16*Rs^2*(Pi + 3/8))/(28*r^2*
  Pi), r)
```

$$\frac{4\left(\pi+\frac{3}{8}\right) Rs^2}{7\,\pi}\,r^{-2}-\frac{1}{4}\,\frac{Rs}{\pi}\,r^{-1}+\frac{-4\,\pi+2}{28\,\pi} \tag{21}$$

```
> Rs := 8.209331978;
```

$$Rs := 8.209331978 \tag{22}$$

```
> evalf(int(4*(Pi + 3/8)*Rs^2/(7*Pi)/r^2 - 1/4*Rs/(Pi*r) + (-4*Pi +
  2)/(28*Pi), r = 0.431 .. 13.79));#The Local Universe comoving
  radial distance is 93 byllion lightyears with initial star at
  431milYears from the big bang;
```

$$93.02203568 \tag{23}$$

```
> evalf(int(4*(Pi + 3/8)*Rs^2/(7*Pi)/r^2 - 1/4*Rs/(Pi*r) + (-4*Pi +
  2)/(28*Pi), r = 0.297 .. 13.79));#Local Universe with the JST 's
  galaxy discocoery JADES-GS-z14&ndash;0 existed only 297 million
  years after the big bang;
```

$$137.8880531 \tag{24}$$

```
> # Thus, comoving radial distance is 137 byllion lightyears of the
  Local Universe;
> evalf(int(4*(Pi + 3/8)*Rs^2/(7*Pi)/r^2 - 1/4*Rs/(Pi*r) + (-4*Pi +
  2)/(28*Pi), r = 0.432 .. 13.4));#The Milky Way
```

$$92.76675873 \tag{25}$$

```
> evalf(int(4*(Pi + 3/8)*Rs^2/(7*Pi)/r^2 - 1/4*Rs/(Pi*r) + (-4*Pi +
  2)/(28*Pi), r = 1.187 .. 13.4));#the comoving radial distance of
  30 BillionLY
```

$$30.04866710 \tag{26}$$

```
> evalf(int(4*(Pi + 3/8)*Rs^2/(7*Pi)/r^2 - 1/4*Rs/(Pi*r) + (-4*Pi +
  2)/(28*Pi), r = 0.929 .. 13.79))
```

$$39.96859485 \tag{27}$$

```
> restart;
> c := 3;
```

$$c := 3 \tag{28}$$

```
> Rs := 8.209331978;
```

$$Rs := 8.209331978 \tag{29}$$

```
> plot(((4 Rs^2 (π + 3/8))/(7 π)) r^(-2) - 1/4 (Rs/π) r^(-1) + (-4 π + 2)/(28 π)), r = 0..2·8.209331978,
  labels = [distance billion lightyears, Λ], filled = [color
  = "blue"], color = red);
```

